



KARNATAKA ICSE SCHOOLS ASSOCIATION

ICSE STD. X Preparatory Examination 2026

Subject – BIOLOGY

Duration: 2 Hrs

Maximum Marks: 80

Date: 21.01.2026

Answers to this Paper must be written on the paper provided separately.

You will not be allowed to write during first 15 minutes.

This time is to be spent in reading the question paper.

The time given at the head of this Paper is the time allowed for writing the answers.

Section A is compulsory. Attempt any four questions from Section B.

The intended marks for questions or parts of questions are given in brackets [].

SECTION A (40 Marks)

(Attempt all questions from this Section)

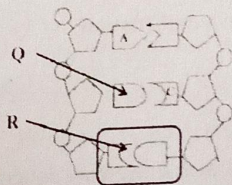
Question 1

Choose the correct answers to the questions from the given options:

[15]

(Do not copy the question, write the correct answers only)

- (i) A cell has five pairs of chromosomes. After mitotic division, the number of chromosomes in the daughter cells will be:
(a) Five (b) Ten (c) Twenty (d) Forty
- (ii) Special anatomical structure through which guttation occurs are:
(a) Hydathode (b) Stomata (c) Lenticel (d) Cuticle
- (iii) The genotype of a person who cannot roll his tongue is _____.
(a) Rr (b) RR (c) rr (d) RRr
- (iv) In the following diagram of a section of a DNA molecule, what is the part labelled Q?



- (a) Cytosine (b) Adenine (c) Thymine (d) Guanine
- (v) Pulse wave is mainly caused by the _____.
(a) Systole of atria (b) Diastole of atria
(c) Systole of the right ventricle (d) Systole of the left ventricle

- (vi) Assertion (A): The sympathetic nervous system is stimulated by the hormone thyroxine.
Reason (R): Sympathetic nervous system prepares the body for violent action against abnormal conditions.
- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A)
(c) (A) is true and (R) is false
(d) (A) is false and (R) is true
- (vii) Which is an example of active transport?
- (a) Carbon dioxide entering a leaf (b) Oxygen moving from the alveoli into the blood
(c) Ion uptake by root hair cells (d) Water uptake by root hair cells
- (viii) Leaf fall occurs when the amount of _____
- (a) Auxin increases (b) Cytokinin decreases (c) Absciscic acid increases (d) Gibberellin decreases
- (ix) Drooping of leaves of Mimosa plant is due to change in:



- (a) Turgor pressure (b) Imbibition (c) Osmosis (d) Diffusion
- (x) Assertion (A): It is stated that population grows at a geometric rate while food production increases at an arithmetic rate.
Reason (R): This difference in growth rates between population and food supply could lead to shortages and famine if population growth is not controlled.
- (a) Both (A) and (R) are true and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(c) (A) is true and (R) is false
(d) (A) is false and (R) is true
- (xi) Which of the following blood vessels in man carries with it the least concentration of urea?
- (a) Hepatic vein (b) Hepatic artery (c) Renal artery (d) Renal vein
- (xii) Which one of the following character is a dominant trait?
- (a) Wrinkled shape of seed (b) Terminal position of flowers
(c) Green colour of seed (d) Axial position of flowers

- xiii) Sita was learning about the human circulatory system. She matched the components with their functions:

Component	Function
P	Transports oxygen in the blood.
Q	Helps in blood clotting.

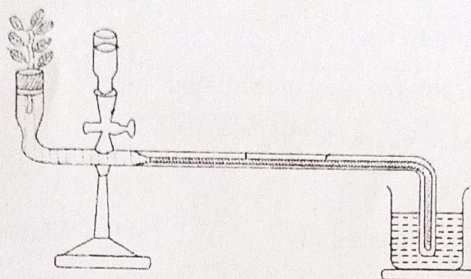
Identify the correct pair.

- (a) P - Red blood cells, Q - Platelets
 (b) P - White blood cells, Q - Plasma
 (c) P - Red blood cells, Q - Plasma
 (d) P - Platelets, Q - White blood cells
- xiv) Assertion (A): The Swachh Bharat Abhiyan' was officially launched on 2nd Oct 2015.
 Reason (R): Our Prime Minister Shri Narendra Modi took the cleanliness pledge at India Gate in Delhi. He wanted to fulfill Mahatma Gandhi's dream of a clean and hygienic India.
- (a) Both (A) and (R) are true
 (b) Both (A) and (R) are false
 (c) (A) is true and (R) is false
 (d) (A) is false and (R) is true
- xv) Which one of the following is the correct route during the transport of sperm in male (human)?
- (a) Vas deferens → Epididymis → Urethra
 (b) Epididymis → Urethra → Vas deferens
 (c) Epididymis → Vas deferens → Urethra
 (d) Urethra → Epididymis → Vas deferens

Question 2

- i) Name the following:
- (a) The hormone which is associated with the 'reabsorption' step of urine production.
 (b) The white portion on the front of the eye.
 (c) Response to a stimulus without intervention of the will of an animal.
 (d) The hormone which stimulates the breakdown of glycogen in liver to glucose.
 (e) The compound formed when haemoglobin combines with carbon dioxide in blood.
- ii) Arrange and rewrite the terms in each group in the correct order so as to be in a logical sequence
- (i) Fibrin, Platelets, Thromboplastin, Fibrinogen, Thrombin.
 (ii) Water molecules, Oxygen, Grana, Photolysis, Photons.
 (iii) Loop of Henle, Bowman's capsule, DCT, Glomerulus, PCT.
 (iv) Ovulation, Parturition, Fertilisation, Gestation, Implantation.
 (v) Endodermis, Cortex, Soil water, Xylem, Root hair.

- (iii) The diagram of an apparatus given below demonstrates a particular process in plants. Study the same and fill in the blanks with suitable correct answer.



- Potometer do not measure the water lost during (a) but measure the water (b) by the cut shoot. They should be made completely (c) and the twig should be cut (d) and under water to avoid (e) of air bubble into the twig.
- (iv) Choose the odd one out from the following terms and name the category to which the others belong: [5]
- (i) Cell wall, Visking bag, Cellophane paper, Parchment paper
 - (ii) Seminal vesicle, Seminiferous tubule, Prostate gland, Cowper's gland
 - (iii) Conjunctiva, Cornea, Cochlea, Choroid
 - (iv) Used Bandages, Pesticides, Face masks, Syringes
 - (v) Simple goitre, Myxoedema, Cretinism, Acromegaly
- (v) Given below is the structure of human ear. Match the structures marked (1) to (6) with their correct function: [5]
- Example:* (1) – (f) converts air currents into vibration.

Structure of human ear	Functions
	(a) equalizes the air pressure
	(b) Dynamic balance
	(c) Amplification of vibration
	(d) Hearing
	(e) Transmit impulses from ear to brain
	(f) Converts air currents into vibration

SECTION B (40 Marks)

(Answer **any four** questions from this section)

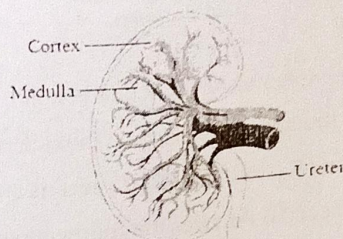
Question 3

- (i) Define the term Diapedesis. [1]
- (ii) Rohan is studying tonicity and how solutes particles affect the tension in a system. He learns about three types of solutions and how they influence the movement of water in and out of cells. [2]

- Duration
- (a) What is the term for the tension developed in a system due to solute particles?
- (b) What type of solution causes water to move into the cell, leading to swell?
- iii) Give scientific reasons:

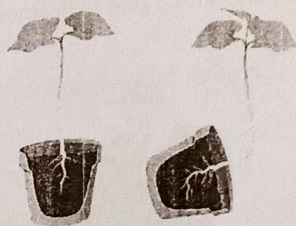
- (a) The children of the same parents, however similar, are different from each other in certain aspects.
- (b) Some women have facial hair like beard and moustache.

- iv) Given below is the longitudinal section of a human kidney. Study the same and answer the following:



- (a) Name the basic structural and functional unit of kidney.
- (b) Why does the medulla of the kidney shows a striped appearance?

- v) Ravi conducts an experiment with a potted plant. Before the plant falls over, he notices certain movements, and after it falls over, he observes different movements. Based on these observations, answer the following questions:



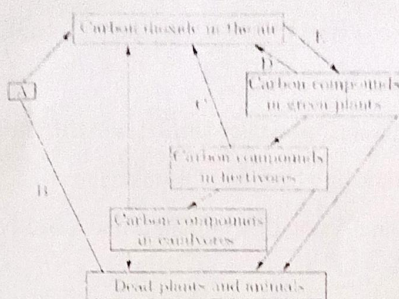
- (a) Name the phenomenon depicted by the shoot in the above diagram.
- (b) Which plant hormone plays an important role in the above movement?
- (c) Complete and rewrite the given statement by filling in the correct terms:

Shoots show positive _____ whereas, roots show positive _____.

Question 4

- i) Define Parturition.
- ii) Give one example of osmosis occurring in human body and a green leaf.
- iii) Differentiate between vasectomy and tubectomy.

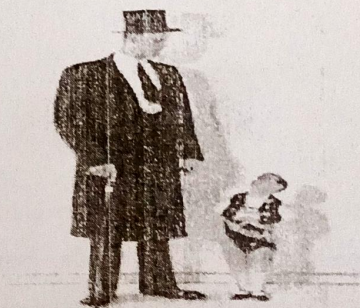
- (iv) The diagram shows how carbon compounds are recycled.



- (a) Name a type of organism you expect to be named in box A.
- (b) State one other way in which the amount of carbon dioxide can be increased.
- (v) A homozygous plant having round (R) and yellow (Y) seed is crossed with another homozygous plant having wrinkled (r) and green (y) seeds. Answer the following questions: [3]
- (a) Give the genotype of the F_1 generation.
- (b) Give the phenotypic ratio of F_2 progeny.
- (c) Name and state the law of purity of gametes.

Question 5

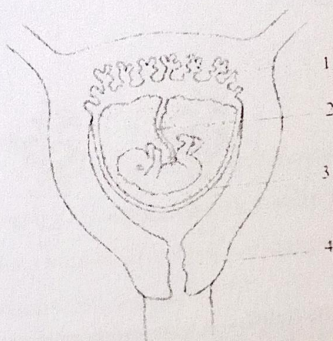
- (i) Foetus can only respire but cannot breathe. Give reason. [1]
- (ii) Given below is the diagram of disorders caused due to hormone. Study the same and answer the following questions: [2]



- (a) Identify the disorders shown in the diagram alongside.
- (b) Name the gland whose hormones cause these disorders with a reason.
- (iii) In an experiment, four freshly plucked leaves (A-D) of a plant, such as those of China rose, was treated as follows: [2]
- (A) coated with Vaseline on its upper surface (B) coated on the lower surface
- (C) coated on both surfaces (D) left uncoated
- All the four leaves (A) to (D) were left in a room for about 24 hours.



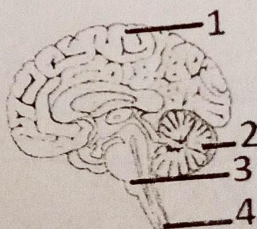
- (a) Which leaf would become most limp? Why? [2]
(b) Which leaf would show least limping? Why? [2]
iv) Draw a neat labelled diagram of a stomatal apparatus. [3]
v) Ravi is studying a diagram related to the development of a foetus. Based on the diagram, answer the following questions:



- (a) Name the parts labelled 1 and 2.
(b) State any one function of the amniotic fluid.
(c) What is the normal gestation period in humans?

Question 6

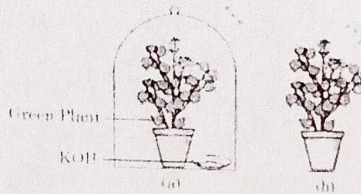
- i) Define the term Power of accommodation. [1]
ii) Differentiate between Menarche and Menopause with reference to its definition. [2]
iii) State the exact location of the following: [2]
(a) Heart
(b) Macula lutea
iv) Draw a neat labelled diagram of a malpighian capsule. [3]
v) The diagram given below represents the human brain. Study the same and answer following questions: [3]



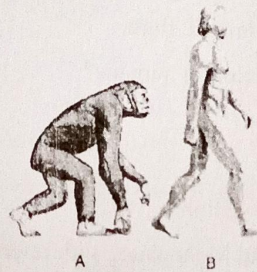
- (a) Compare the location of cyton and axon in part 1 and part 4.
(b) Why is injury to part 3 cause immediate death of a person?
(c) Name the collective term for the protective membranes covering the brain.

Question 7

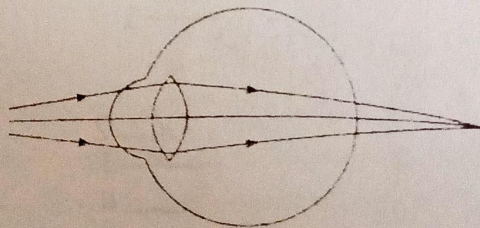
- (i) Abbreviate the term NADP. [1]
- (ii) Differentiate between Diabetes mellitus and Diabetes insipidus based on the hormones involved. [1]
- (iii) In a biology experiment, two healthy green plants were placed in the dark for 24 hours. They were then set up as shown in the figure and left for four hours. Then a leaf was taken from each plant and the chlorophyll was removed from the leaves. [2]



- (a) How is chlorophyll removed? [1]
- (b) What is the main aim of this experiment? [1]
- (iv) The figure shows a human ancestor. Study the same and answer the following. [2]



- (a) Mention any *two* contrasting characters between the two stages. [1]
- (b) Write all the stages of human evolution in their correct sequence. [1]
- (v) Given below is a diagram depicting a defect of the human eye. Study the same and then answer the [3]
- questions that follow:



- (a) Identify the defect of the eye by mentioning the technical term. [1]
- (b) Mention one reason for this defect. [1]
- (c) Name the type of lens used to correct this defect. [1]

Section 8

Define Population Density.

At the start of a race, athletes react to a starter gun. This reaction involves a number of events. [1]

(1) Co-ordination by brain [2]

(2) Detection by receptors

(3) Stimulus

(4) Response

(5) Effector change

(a) Arrange the above mentioned events in correct order.

(b) Write one difference between voluntary and reflex action.

iii) The picture given below shows a kind of effects of various pollution on climate and environment. Study the same and answer the questions that follow: [2]

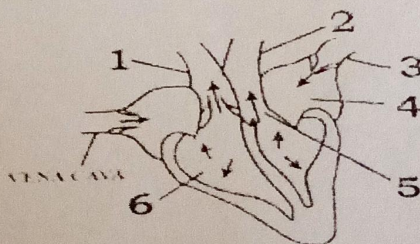


(a) State the kind of effect shown in the above picture.

(b) Mention any one harmful effect of this pollution on human health.

iv) State the difference between Corpus collasum and Corpus luteum based on their location. [2]

v) The given diagram represents one of phase of the heart. Study the given diagram carefully and answer the questions that follow: [3]



(a) Identify the phase. Give a reason for your choice.

(b) What type of blood flows through part 1 and part 2?

(c) The left ventricle has thicker walls than the right ventricle. Give reason.